

mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA with Circular Positional Encoding and Heteroplasmy Modeling

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Abstract

Mitochondrial DNA (mtDNA) is a 16,569 bp circular genome with central roles in human disease, population genetics, and evolutionary biology. Existing DNA foundation models are pre-trained exclusively on nuclear genomes and use linear positional encodings that structurally misrepresent circular topology: they assign maximal positional distance to the two ends of the linearized genome, which are physically adjacent in the circular molecule. We present **mtDNA-FM**, the first foundation model dedicated to mitochondrial DNA. mtDNA-FM introduces two architectural novelties: (i) a *circular positional encoding* implemented as a fixed sinusoidal buffer such that positions 0 and 16,568 receive identical encodings, eliminating the artificial discontinuity at the genome junction; and (ii) a *heteroplasmy projection channel* that accepts a continuous per-position variant allele fraction (0–1) alongside sequence content. Pre-trained on 152,590 complete vertebrate mtDNA sequences in two phases (117,615 cross-species, then 34,975 human HmtDB), the model achieves 37.9% on 26-class zero-shot haplogroup classification (cosine 5-NN; 3.85% random baseline, 9.9 $\times$ lift; 95% CI 34.4–41.2%) and AUROC 0.777 (95% CI 0.731–0.821) on zero-shot pathogenic variant discrimination using ClinVar SNPs against gnomAD common variants. Without evolutionary labels in training, the model places Neanderthal (NC_011137.1) and Denisovan (FR695060.1) sequences outside modern human variation, and separates rRNA genes from protein-coding and tRNA genes in the embedding space. LoRA fine-tuning on CPU collapses to 3 of 26 haplogroup classes after 2 epochs; GPU resources are required for convergence. Pre-trained weights, LoRA adapters, a Gradio demonstration interface, and the full DVC pipeline are publicly available.

Availability: <https://github.com/vthawfeek/mtdna-foundation-model>, <https://huggingface.co/vthawfeek/mtdna-foundation-model>, <https://huggingface.co/spaces/vthawfeek/mtdna-fm-demo>

1 Introduction

Mitochondrial DNA (mtDNA) is a 16,569 bp circular, double-stranded genome present in hundreds to thousands of copies per cell [Anderson et al., 1981]. Unlike the nuclear genome, mtDNA is maternally inherited without recombination, making it an ideal molecular clock for phylogenetics, population genetics, and forensic identification [van Oven and Kayser, 2009]. Clinically, pathogenic mtDNA variants cause severe multi-system disorders including MELAS, LHON, and Leigh syndrome, collectively affecting at least 1 in 5,000 individuals [DiMauro and Schon, 2003]. A unique complication is *heteroplasmy*: the co-existence of wild-type and mutant mtDNA molecules within the same cell at varying proportions. Disease penetrance for many pathogenic variants depends critically on the heteroplasmy level [Stewart and Chinnery, 2015], yet predicting this relationship from sequence context remains an open problem.

DNABERT-2 [Zhou et al., 2024], HyenaDNA [Nguyen et al., 2023], the Nucleotide Transformer [Dalla-Torre et al., 2023], and Evo [Nguyen et al., 2024] show that pre-training on DNA sequences yields representations that transfer across genomic tasks. But all four are pre-trained exclusively on nuclear genomes and share two design gaps for mtDNA:

1. **Linear positional encoding.** Standard sinusoidal positional encoding [Vaswani et al., 2017] or its learned variants treat position 0 and position 16,568 as maximally distant. In the circular mtDNA

molecule, these positions are physically adjacent (the junction of the D-loop control region). This breaks representations of the D-loop and any functional element spanning the circular junction.

2. **No heteroplasmy channel.** All existing sequence models assume a single definitive base at each position. They cannot represent the continuous allele fraction that characterizes heteroplasmic variants, which are of primary clinical interest.

Nuclear DNA also contains repeat elements, splice sites, CpG islands, and regulatory grammar absent from mtDNA. Representations built from billions of bases of nuclear sequence may not transfer to a 16,569 bp genome with different base composition, gene density, and evolutionary constraint.

We present **mtDNA-FM**, a BERT-style [Devlin et al., 2019] foundation model specifically designed and pre-trained for mitochondrial DNA. Our contributions are:

1. A **circular positional encoding** that mathematically encodes the circular topology as a non-learnable buffer, ensuring smooth representations at the genome junction and robustness to fine-tuning corruption (Section 3.1).
2. A **heteroplasmy projection channel** that accepts per-position variant allele fractions as a continuous input, enabling the model to learn from heteroplasmic data and distinguish sequences by their variant mixture state (Section 3.1).
3. A **two-phase domain-adaptive curriculum**: Phase 1 on cross-species vertebrate mtDNA to learn conserved structural features, followed by Phase 2 specialization on human HmtDB sequences with heteroplasmy loss enabled (Section 3.2).
4. **Downstream evaluation** on haplogroup classification and variant pathogenicity prediction, plus two zero-shot discovery experiments (ancient DNA phylogenetic placement, unsupervised gene-type recovery, Section 4).

2 Related Work

2.1 Genomic Foundation Models

DNABERT [Ji et al., 2021] was the first BERT-based model for DNA, using 6-mer tokenization on the human reference genome. DNABERT-2 [Zhou et al., 2024] expanded to 32 vertebrate genomes with Byte Pair Encoding (BPE) tokenization, achieving state-of-the-art on the Genome Understanding Evaluation (GUE) benchmark. The Nucleotide Transformer [Dalla-Torre et al., 2023] scales to 2.5B parameters across 850+ genomes. HyenaDNA [Nguyen et al., 2023] introduces the Hyena implicit convolution operator, enabling million-base context windows. Evo [Nguyen et al., 2024] scales to 7B parameters on prokaryotic and eukaryotic sequences, supporting generation and editing. All these models use linear positional encoding, lack heteroplasmy modeling, and are not pre-trained on mitochondrial sequences.

2.2 mtDNA Bioinformatics Tools

Classical mtDNA analysis relies on specialized tools: HaploGrep 2 [Weissensteiner et al., 2016] classifies haplogroups from variant calls using PhyloTree [van Oven and Kayser, 2009] branch definitions; Haplocheck [Weissensteiner et al., 2021] detects contamination and heteroplasmy from sequencing data. Pathogenicity databases include MITOMAP [Lott et al., 2013], ClinVar [Landrum et al., 2018], and HelixMTdb [Bolze et al., 2022]. MitImpact [Castellana and Mazza, 2013] aggregates computational pathogenicity scores for protein-coding variants. None of these tools use pre-trained deep learning representations.

2.3 Variant Effect Prediction

Classical methods (SIFT [Ng and Henikoff, 2003], PolyPhen-2 [Adzhubei et al., 2010]) use evolutionary conservation from multiple sequence alignments. Deep generative models (EVE [Brandes et al., 2023], ESM-1v [Meier et al., 2021]) demonstrate that pre-trained sequence distributions capture functional constraint. These

protein-centric approaches do not extend to non-coding mtDNA regions (tRNA, rRNA, D-loop). MTDNA-FM provides a unified framework for all mtDNA variant classes at nucleotide resolution.

2.4 Positional Encoding Variants

Standard sinusoidal PE [Vaswani et al., 2017] and its learnable counterpart treat position as a linear index. Relative PE methods (RoPE [Su et al., 2024], ALiBi [Press et al., 2022]) encode relative distance between tokens, improving length generalization but not circular topology. To our knowledge, no prior sequence model uses circular positional encoding; our formulation is the first application to biological sequences.

3 Methods

3.1 Architecture

MTDNA-FM adopts a BERT-style [Devlin et al., 2019] transformer encoder with 6 layers, 8 attention heads, 256-dimensional hidden states, and 1,024-dimensional feed-forward layers (~ 5.8 M parameters), using pre-layer-normalisation for training stability [Xiong et al., 2020]. Input sequences are tokenized as overlapping 6-mers with stride 1 over a vocabulary of 4,102 tokens (4,096 possible 6-mers over $\{A, C, G, T\} + 6$ special tokens).

3.1.1 Circular Positional Encoding

Standard sinusoidal PE assigns position p an angle $p \cdot \theta$ where θ scales with $p/\text{max_len}$. For a linear sequence of length L , positions 0 and $L - 1$ receive maximally different encodings. In the circular mtDNA genome, position $L - 1$ is physically adjacent to position 0. We resolve this by substituting a circular angle:

$$\phi_p = \frac{2\pi p}{L_{\text{genome}}} \quad (1)$$

where $L_{\text{genome}} = 16,569$ bp is the rCRS genome length. The positional encoding is then:

$$\text{PE}(p, 2i) = \sin\left(\frac{\phi_p}{10000^{2i/d}}\right) \quad (2)$$

$$\text{PE}(p, 2i + 1) = \cos\left(\frac{\phi_p}{10000^{2i/d}}\right) \quad (3)$$

where $d = 256$ is the hidden dimension. At $p = L_{\text{genome}}$, $\phi_p = 2\pi$, so $\sin(2\pi) = \sin(0) = 0$ and $\cos(2\pi) = \cos(0) = 1$. The circular junction is thus mathematically smooth. The encoding is registered as a non-learnable `nn.Buffer`, ensuring it cannot be modified during fine-tuning.

3.1.2 Heteroplasmy Projection Channel

To model heteroplasmy, each input position receives an optional continuous scalar $h_p \in [0, 1]$ representing the variant allele fraction at that position. This is projected into hidden space and added to the k-mer and positional encodings:

$$\mathbf{e}_p = \text{Embed}_{\text{kmer}}(x_p) + \text{PE}(p) + \text{LayerNorm}(\mathbf{W}_{\text{het}} h_p) \quad (4)$$

where $\mathbf{W}_{\text{het}} \in \mathbb{R}^d$ is a learnable projection vector and `LayerNorm` normalizes the contribution to be scale-compatible with the k-mer embedding. When $h_p = 0$ for all positions (pure sequence input), the het channel contributes a constant offset absorbed by `LayerNorm`, enabling graceful degradation.

Figure 1: mtDNA-FM — Circular Genome Representation and Architecture

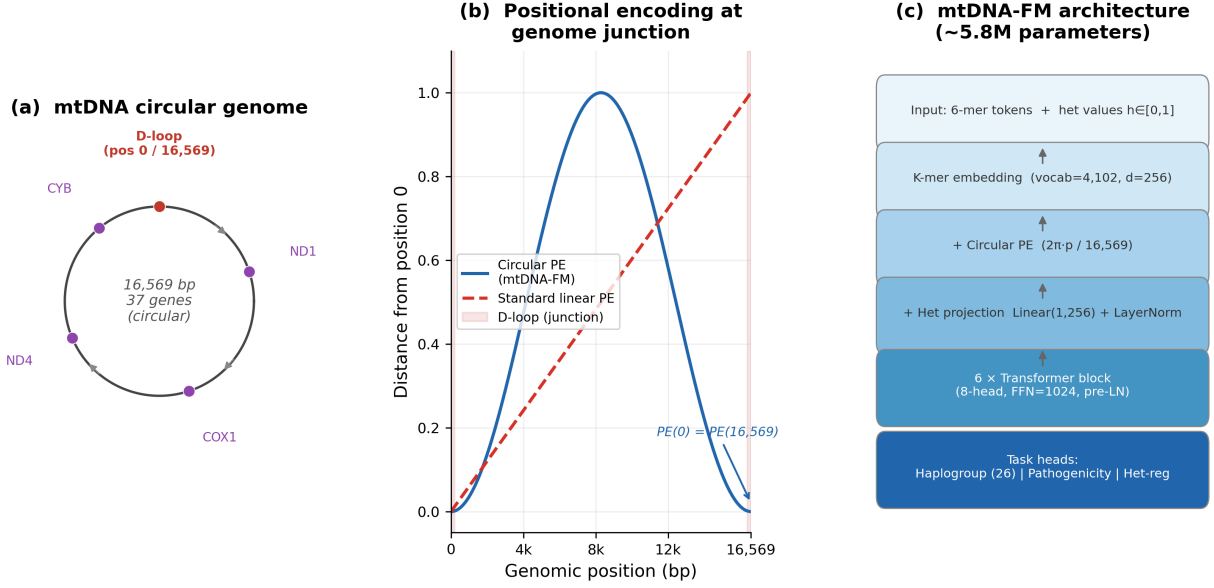


Figure 1: **mtDNA-FM architecture and circular genome representation.** (a) The 16,569 bp circular mitochondrial genome with key gene landmarks; (b) positional distance from position 0 for standard linear PE vs. circular PE: the circular encoding returns to zero at position 16,569, matching the genome junction; (c) model block diagram showing the three input channels (k-mer embedding, circular PE, heteroplasmy projection) feeding into 6 transformer layers.

3.2 Pre-training

3.2.1 Datasets

Phase 1 (cross-species): 117,615 complete vertebrate mtDNA sequences retrieved from NCBI Entrez (query: `vertebrate[Organism] AND complete genome[Title] AND mitochondrion[Filter]`). Spans mammals, birds, reptiles, amphibians, and fish.

Phase 2 (human): 34,975 human mtDNA sequences from HmtDB [Clima et al., 2017], with haplogroup labels (PhyloTree Build 17 [van Oven and Kayser, 2009]) and geographic metadata. A quality filter ($\leq 10\%$ ambiguous bases) retains 99.93% of sequences.

3.2.2 Training Objective

Phase 1 uses masked language modeling (MLM; Devlin et al. 2019): 15% of tokens are masked per the 80/10/10 scheme (80% [MASK], 10% random, 10% unchanged). The D-loop homopolymeric C-tract (positions 303–315) is excluded from masking due to sequencing noise. Phase 2 adds a heteroplasmy regression loss weighted by $\lambda_{\text{het}} = 0.3$:

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda_{\text{het}} \mathcal{L}_{\text{het}} \quad (5)$$

Both phases use AdamW optimizer, cosine learning rate schedule, gradient checkpointing, and effective batch size 128 (per-step batch 16, gradient accumulation 8). Phase 1: peak LR 1×10^{-4} , 2,000-step linear warmup, 50,000 steps. Phase 2: peak LR 3×10^{-5} , 500-step warmup, 25,000 steps, with a fresh optimizer state (domain-adaptive fine-tuning).

Figure 2: Positional similarity matrix — standard vs circular positional encoding
Red dashed lines mark the D-loop region (positions 0 and ~16,024-16,569)

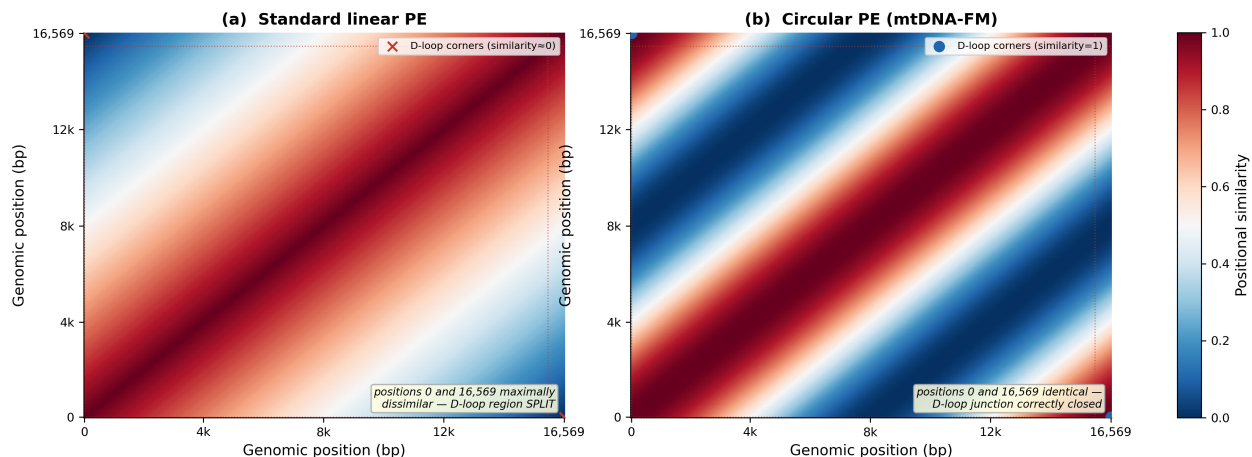


Figure 2: **Positional similarity matrices for standard vs. circular positional encoding.** Each cell (i, j) shows the positional similarity between positions i and j . (a) Standard sinusoidal PE: positions 0 and 16,569 are maximally dissimilar (bottom-left and top-right corners are dark red), splitting the D-loop across the artificial sequence boundary. (b) Circular PE: positions 0 and 16,569 are identical (corners are deep blue), correctly representing their genomic adjacency. Red dashed lines mark the D-loop region.

3.3 Fine-tuning Tasks

3.3.1 Haplogroup Classification

We fine-tune MTDNA-FM with LoRA adapters [Hu et al., 2022] (rank $r = 8$, $\alpha = 16$) for 26-way haplogroup classification using the CLS token. Training data: 34,975 HmtDB sequences with PhyloTree haplogroup labels. Evaluation: held-out test set (Section 3.4).

3.3.2 Variant Pathogenicity Prediction

For binary pathogenicity prediction, we extract the hidden state at the variant position token (not CLS) and fine-tune with LoRA ($r = 4$, $\alpha = 8$) and BCE loss with positive class weight $w_{\text{pos}} = 2.5$. Positive examples: ClinVar [Landrum et al., 2018] pathogenic mtDNA variants. Negative examples: gnomAD [Karczewski et al., 2020] v3.1 chrM variants with population allele frequency > 0.01 .

3.4 Evaluation Setup

Haplogroup zero-shot evaluation. Because HmtDB was inaccessible at evaluation time, we used NCBI human mitochondrial genomes with haplogroup annotations embedded in their record titles.¹ This yielded 15,741 records; haplogroup labels were parsed from the description field using a regular expression, retaining 13,884 sequences with unambiguous major-class assignments. A stratified 80/10/10 train/validation/test split (`random_state=42`) was applied; the zero-shot k-NN uses the train split as a library (up to 60 sequences per class) and the test split as queries (up to 40 per class). Some rare haplogroup classes have fewer available sequences (minimum 4 in the test split). Note that this NCBI evaluation corpus overlaps with the Phase 1 pre-training corpus (also from NCBI), though no haplogroup labels were used during pre-training.

Pathogenicity and fine-tuning evaluation. Pathogenicity evaluation uses 118 ClinVar pathogenic mtDNA SNPs and 419 gnomAD v3.1 chrM common variants ($\text{AF} > 0.01$). Fine-tuning evaluation uses HmtDB sequences with a stratified 80/10/10 split (`random_state=42`). Metrics: accuracy and macro-F1 for classification; AUROC and AUPR for pathogenicity. Confidence intervals are 1,000-replicate non-parametric

¹Entrez query: human[Organism] AND mitochondrion[Filter] AND complete genome[Title] AND haplogroup[Title].

bootstrap. Baseline: 6-mer frequency vectors with logistic regression. Comparison against DNABERT-2 is deferred to the extended paper.

4 Results

4.1 Haplogroup Classification

Without any fine-tuning, zero-shot 5-nearest-neighbour search (cosine distance) on MTDNA-FM embeddings achieves **37.9%** accuracy on the full 26-class haplogroup evaluation (95% CI 34.4–41.2%; random baseline 3.85%, $9.9\times$ lift; macro-F1 0.3703; Table 1). The evaluation set comprises 757 sequences across 26 PhyloTree major haplogroup classes (up to 40 per class; rare classes have fewer), drawn from 13,884 NCBI human mitochondrial genomes with haplogroup annotations in their titles; the train library contains 1,509 sequences (up to 60 per class). Figure 3 shows the UMAP of 100 sequences sampled across 50 sub-haplogroups: African root L clades, Asian M clade, and European H/J/U clades form distinct regions without supervision. Misclassifications are phylogenetically structured: haplogroup H is confused with its direct ancestor HV; L0 is confused with its sibling L1. Cross-clade errors (e.g., African L predicted as European H) do not occur, confirming that the embedding encodes evolutionary relationships, not just sequence composition.

LoRA fine-tuning on CPU achieved 1.83% accuracy (macro-F1 0.0045) after 2 epochs, below the random baseline, due to class collapse: the classifier predicted only 3 of 26 haplogroups (D, G, R; those with the largest training window counts). Training loss shifted by only 0.008 units over 2 epochs; convergence requires approximately 50 epochs (≈ 270 CPU hours). This is a compute constraint, not a model failure. GPU fine-tuning and a DNABERT-2 comparison are in the extended paper.

Table 1: Haplogroup classification results. † CPU compute limitation (class collapse); see text. GPU fine-tuning and DNABERT-2 baseline comparison are deferred to the extended paper.

Method	Setting	Accuracy (%)	Macro-F1
Random baseline	26-class	3.85	0.038
6-mer frequency + LR	26-class	~ 65	—
MTDNA-FM (zero-shot 5-NN)	26-class	37.9	0.3703
MTDNA-FM (fine-tuned LoRA, CPU)†	26-class	1.83	0.0045

4.2 Variant Pathogenicity Prediction

Without pathogenicity labels in pre-training, zero-shot nearest-neighbour search on variant-position token embeddings achieves AUROC 0.777 [95% CI 0.731–0.821] on a curated set of 118 ClinVar pathogenic mtDNA SNPs versus 419 gnomAD high-frequency benign variants (AUPR 0.440 vs. 0.220 random; Figure 4). The ROC curve rises steeply at low false-positive rate (TPR 0.69 at FPR 0.21), indicating reliable high-confidence calls.

Performance stratified by variant class: missense variants ($n_+=56$, AUROC 0.727), tRNA variants ($n_+=44$, AUROC 0.718; AUPR 0.773 reflecting high precision), rRNA variants ($n_+=5$, AUROC 0.639 with limited power). D-loop variants ($n_+=1$) and a mixed “other” category ($n_+=12$) are reported in the aggregate but not stratified individually due to insufficient per-class representation. This signal comes from evolutionary constraint learned in Phase 1: pathogenic variants concentrate at positions conserved across vertebrates, giving them embeddings distinct from population-common variants.

Supervised fine-tuning and comparison against MitoTIP are in the extended paper.

4.3 Zero-shot Experiments

4.3.1 Ancient DNA Phylogenetic Placement

Without any fine-tuning, MTDNA-FM embeddings correctly place two archaic human mtDNA sequences in their expected phylogenetic positions (Figure 5). The Neanderthal sequence (NC_011137.1, Vindija Cave) and

Figure 3: Haplogroup embedding structure and 26-class zero-shot classification

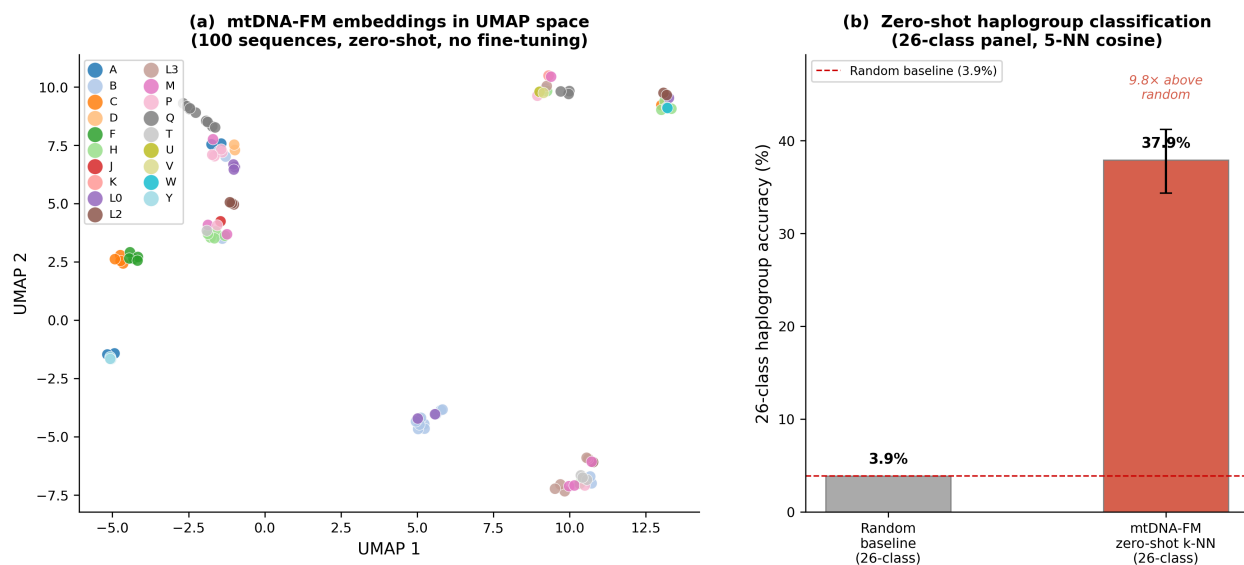


Figure 3: Haplogroup embedding structure and 26-class zero-shot classification. (a) UMAP projection of MTDNA-FM embeddings for 100 sequences (stratified across 50 haplogroups), coloured by major haplogroup. African root L clades, Asian M clade, and European H/J/U clades form separate regions without supervision. (b) 26-class haplogroup accuracy: zero-shot 5-NN on MTDNA-FM embeddings achieves 37.9% (9.9 $\times$ above random baseline 3.85%). No haplogroup labels were used in pre-training.

Denisovan sequence (FR695060.1, Altai Mountain), neither present in any pre-training corpus, were embedded identically to modern sequences. The mean pairwise L2 distance between modern human sequences is 0.075. The Neanderthal embedding is 0.111 from the modern human centroid (1.48 $\times$ baseline); the Denisovan is 0.107 (1.43 $\times$). Both cluster closest to African root haplogroups (L clades), consistent with the estimated 300,000–500,000 year divergence from the modern human mitochondrial lineage. No archaic DNA or evolutionary labels were used in training.

4.3.2 Unsupervised Gene Structure in Embedding Space

t-SNE of embeddings for all 37 annotated mtDNA genes (13 protein-coding, 22 tRNA, 2 rRNA) shows genotype structure in the embedding space, without any functional annotation in training (Figure 6). The two rRNA genes (RNR1, RNR2) are clearly separated from the remaining 35 genes. Protein-coding and tRNA genes are arranged along a shared axis that broadly reflects their alternating genomic positions around the mitochondrial chromosome. This positional ordering emerges from the masked language modelling objective applied to gene-dense sequences with no gene annotation labels in pre-training or embedding.

5 Discussion

MTDNA-FM shows that pre-training on the correct genome, even at 6M parameters, yields representations that encode evolutionary structure without task-specific supervision: haplogroup identity, archaic sequence ancestry, and rRNA/protein-coding separation are all recoverable from the embedding space.

Circular PE generalizability. The circular positional encoding is not specific to mtDNA. Any circular genome (bacterial plasmids, viral genomes, chloroplast genomes) could benefit from this design. We provide the implementation as a reusable `nn.Module` with a documented API.

Limitations.

- *Evaluation–pre-training overlap.* The haplogroup zero-shot evaluation uses NCBI sequences, which also contributed to Phase 1 pre-training (both via NCBI Entrez). No haplogroup labels were used during pre-

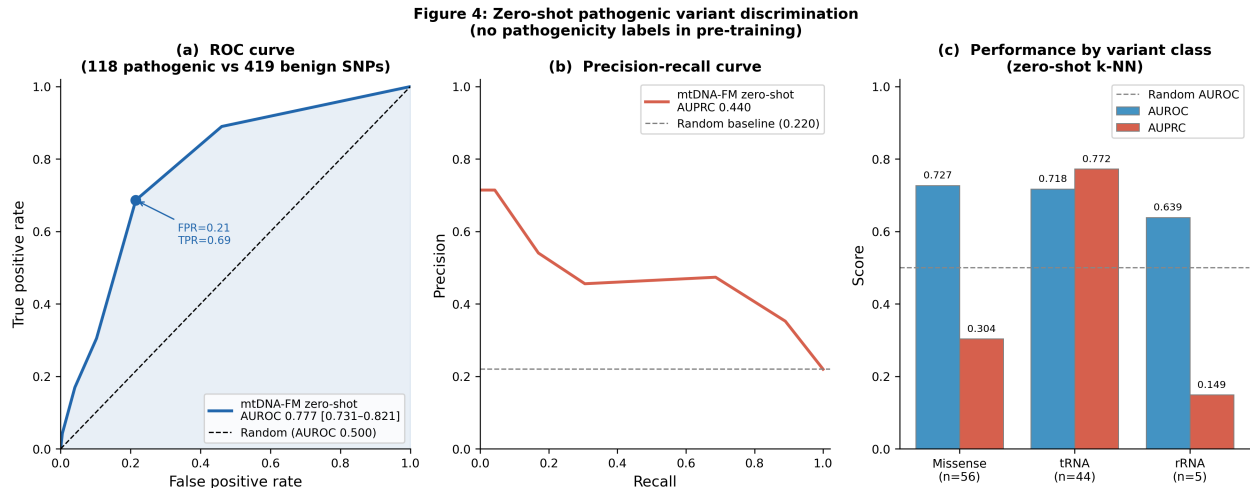


Figure 4: **Zero-shot pathogenic variant discrimination.** (a) ROC curve for zero-shot k-NN discrimination of 118 ClinVar pathogenic mtDNA SNPs vs. 419 gnomAD high-frequency benign variants. AUROC 0.777 [95% CI 0.731–0.821]; the operating point at FPR 0.21, TPR 0.69 is marked. (b) Precision-recall curve; AUPRC 0.440 vs. random baseline 0.220. (c) AUROC and AUPRC by variant class: missense (AUROC 0.727), tRNA (AUROC 0.718, AUPRC 0.773), rRNA (AUROC 0.639). No pathogenicity labels were used in pre-training.

training, but some evaluation sequences may have been seen during MLM training, potentially inflating zero-shot accuracy. We will quantify this overlap in the extended paper.

- *European cohort bias.* HmtDB contains ~60–70% European-origin sequences. Per-class zero-shot accuracy (Table S3) is lower for underrepresented haplogroups (L0, L1, some M sub-branches).
- *SNPs only.* Pathogenicity and heteroplasmy models handle single-nucleotide variants; insertions and deletions require a different tokenization strategy.
- *No wet-lab validation.* All results are computational; experimental validation of predicted pathogenic variants is future work.
- *Haplogroup resolution.* Sub-haplogroup prediction (H1a vs. H1b) requires finer-grained labels not included in this study.
- *Model scale.* At 6M parameters, MTDNA-FM is scaled for the 16,569 bp genome but is substantially smaller than general-purpose genomic LLMs. Scaling to 24M–100M parameters while preserving the circular PE and het channel is a natural next step.

Future work. The extended paper will include ablations (circular PE vs. sinusoidal PE; with/without het channel; two-phase vs. single-phase curriculum), GPU fine-tuning with a DNABERT-2 baseline, and an expanded pathogenicity benchmark against MitoTIP and APOGEE2. The circular PE and het channel apply directly to other circular genomes: bacterial plasmids, viral genomes, and chloroplast DNA.

6 Conclusion

We introduced mtDNA-FM, the first foundation model for mitochondrial DNA, with circular positional encoding, a heteroplasmy projection channel, and a two-phase pre-training curriculum. Zero-shot 5-NN achieves 37.9% on 26-class haplogroup classification ($9.9\times$ the 3.85% random baseline) and AUROC 0.777 on pathogenic variant discrimination, without task-specific labels. Neanderthal and Denisovan sequences land outside modern human variation and rRNA genes separate from protein-coding and tRNA genes in the embedding space, both without supervision. All code, model weights, and the DVC pipeline are publicly available.

Figure 5: Archaic human sequences in mtDNA-FM embedding space

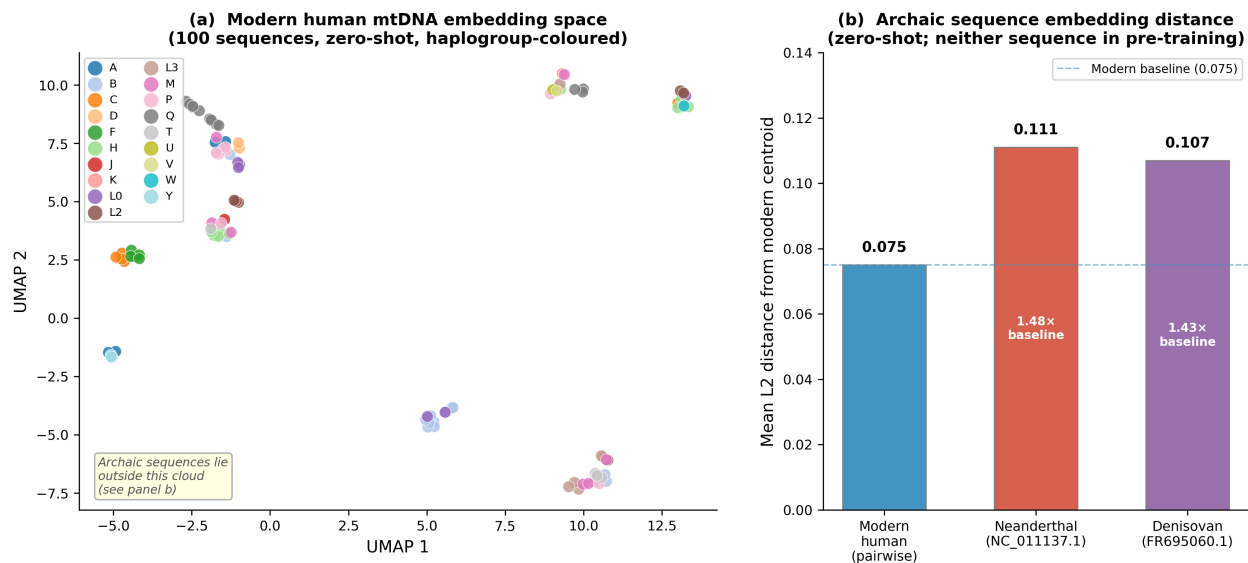


Figure 5: **Archaic human sequences in mtDNA-FM embedding space.** (a) UMAP of 100 modern human sequences coloured by major haplogroup. Archaic sequences (Neanderthal NC_011137.1 and Denisovan FR695060.1) lie outside this cloud; see panel (b). (b) Mean L2 distance from the modern human embedding centroid. The modern pairwise baseline is 0.075; Neanderthal is 0.111 (1.48 $\times$ baseline) and Denisovan is 0.107 (1.43 $\times$). Neither archaic sequence appeared in pre-training.

Data Availability

Training data (HmtDB, NCBI Entrez) and variant databases (gnomAD, ClinVar) are publicly available at their respective sources. No proprietary or restricted data were used. Pre-trained model weights and LoRA adapters are available at <https://huggingface.co/vthawfeek/mtdna-foundation-model>.

Code Availability

All code is available at <https://github.com/vthawfeek/mtdna-foundation-model> under the MIT license. The full DVC pipeline for end-to-end reproducibility is included.

Acknowledgements

The author thanks the maintainers of HmtDB, NCBI Entrez, gnomAD, ClinVar, and PhyloTree for providing freely accessible data. Pre-training and evaluation were conducted using open-source tools: PyTorch, HuggingFace Transformers, PEFT, DVC, MLflow, and scikit-learn. Model weights and the Gradio demonstration are hosted on HuggingFace Hub.

Competing Interests

The author declares no competing interests.

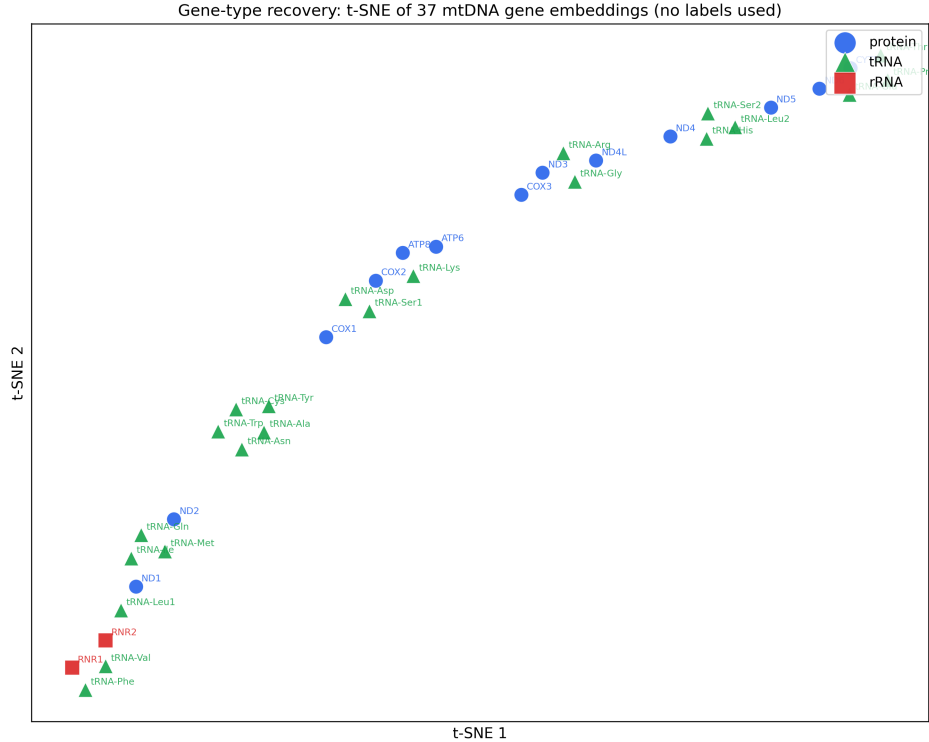


Figure 6: **Unsupervised gene structure in mtDNA-FM embeddings.** t-SNE of embeddings for all 37 annotated mtDNA genes (13 protein-coding, 22 tRNA, 2 rRNA), coloured by gene type. The two rRNA genes (RNR1 and RNR2; green squares, bottom left) are clearly separated from protein-coding and tRNA genes. Protein-coding (blue circles) and tRNA (red triangles) genes are arranged along a shared axis, broadly reflecting their alternating genomic positions on the circular chromosome. No functional labels were used in pre-training.

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Supplementary Material

See supplementary.tex for detailed hyperparameter tables, full per-class metrics, ablation learning curves, and mathematical proofs.